LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 4 Test (Set B): Carbon and its Compounds

Chapter / Topic: Carbon and its Compounds

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SECTION A

Q1. The next homologue of methane (CH_4) is: **[1]**

Answer: (c) C_2H_6 (ethane)

Q2. The functional group present in propanal is: **[1]**

Answer: (d) aldehyde (–CHO)

Q3. The two carbon atoms in ethene are joined by: **[1]**

Answer: (b) a double bond

Q4. Soaps are: **[1]**

Answer: (c) sodium or potassium salts of long-chain carboxylic acids

Q5. Vegetable ghee is made from vegetable oils by: **[1]**

Answer: (a) hydrogenation (with a nickel catalyst)

Q6. **Assertion (A):** Pure ethanoic acid is called glacial acetic acid. **Reason (R):** Pure ethanoic acid freezes in cold climates because its melting point is… **[1]**

Answer: (a)

SECTION B

Q7. Why do covalent compounds generally have low melting and boiling points? Why are they poor conductors of electricity? **[2]**

Answer: The forces between their molecules (intermolecular forces) are weak, so little energy is needed to separate them. They are poor conductors because the bonds are formed by sharing electrons, so no ions are formed.

- Weak intermolecular forces: 1
- No ions: 1

OR, answer: (a) Bromoethane (b) Methanal (c) Propanone (d) Propanol

- ½ each

Q8. What is a homologous series? Give two of its characteristics. **[2]**

Answer: A series of compounds with the same functional group, in which successive members differ by a $–CH_2–$ unit. Characteristics (any two): successive members differ by 14 u in molecular mass; they have similar chemical properties; their physical properties (such as melting and boiling points) change gradually; they can be represented by one general formula.

- Definition: 1
- Two characteristics: ½ + ½

SECTION C

Q9. Answer the following: **[3]**

(i) C_nH_{2n} [1]
(ii) Propane, C_3H_8 [1]
(iii) 14 u (one $–CH_2–$ unit) [1]

Q10. What is a substitution reaction? Give a balanced equation and the condition needed. **[3]**

Answer: A reaction in which an atom or group of atoms in a molecule is replaced by another atom or group. Methane reacts with chlorine in the presence of sunlight: $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$. A hydrogen atom is replaced by a chlorine atom.

- Definition: 1
- Equation: 1
- Sunlight: 1

OR, answer: A soap molecule has a hydrophilic ionic end and a long hydrophobic hydrocarbon tail. In water, the tails stay away from the water by clustering together inside, with the ionic ends on the outside, which forms a micelle. In ethanol, no micelle forms, because the hydrocarbon tail dissolves in ethanol and there is no need for the tails to cluster.

- Two ends of soap molecule: 1
- Micelle in water: 1
- No micelle in ethanol, with reason: 1

SECTION D

Q11. Answer the following about ethanoic acid: **[5]**

(i) (a) $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$ (b) $2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$ (c) $CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$ [3]

- Each equation: 1

(ii) Ethanoic acid: as vinegar (5–8% solution), used as a preservative in pickles. Esters: in perfumes and as flavouring agents. [2]

- Each use: 1

OR, answer: (a) Catenation is carbon's ability to form bonds with other carbon atoms, giving long chains, branches and rings. Tetravalency means carbon has a valency of four, so it can bond with four other atoms. (b) Ethane: $CH_3–CH_3$. Ethene: $CH_2=CH_2$. Ethyne: $HC\equiv CH$.

- (a) 1 + 1
- (b) 1 each

SECTION E

Q12. Hard water contains salts of calcium and magnesium. When soap is used in hard water, it reacts with these salts to form an insoluble substance called… **[4]**

(i) Calcium and magnesium ions. [1]

(ii) An insoluble precipitate formed when soap reacts with the calcium and magnesium salts in hard water. [1]

(iii) Detergents do not form scum with calcium and magnesium ions, so they lather well and less is wasted. Uses: shampoos, and products for washing clothes. [2]

- No scum: 1
- Two uses: 1